Analysis of GDP per capita vs life expectancy for the year 2020 for Nigeria, Ghana and Uganda.

Report Developed by: Edikan Uwatt **Date: 11th of October 2026**

1. GDP per Capita and Life Expectancy, 2020

Focus countries: Nigeria, Ghana and Uganda. Data: Gapminder (gdp_pcap.csv, lex.csv). 193 countries with data for both indicators in 2020.

1.1. 1. Summary

In 2020, the two indicators are strongly and positively related: the correlation between log GDP per capita and life expectancy is r = 0.83 (Spearman rank correlation on the raw values = 0.84). On average, a tenfold increase in GDP per capita is associated with about 11.7 more years of life expectancy.

All three focus countries sit below the world median on both indicators (median GDP per capita $14,397; median life expectancy 72.6 yrs). Ghana has the highest income of the three, but Uganda has the highest life expectancy despite having less than half of Ghana's GDP per capita.

1.2. 2. GDP per capita, 2020

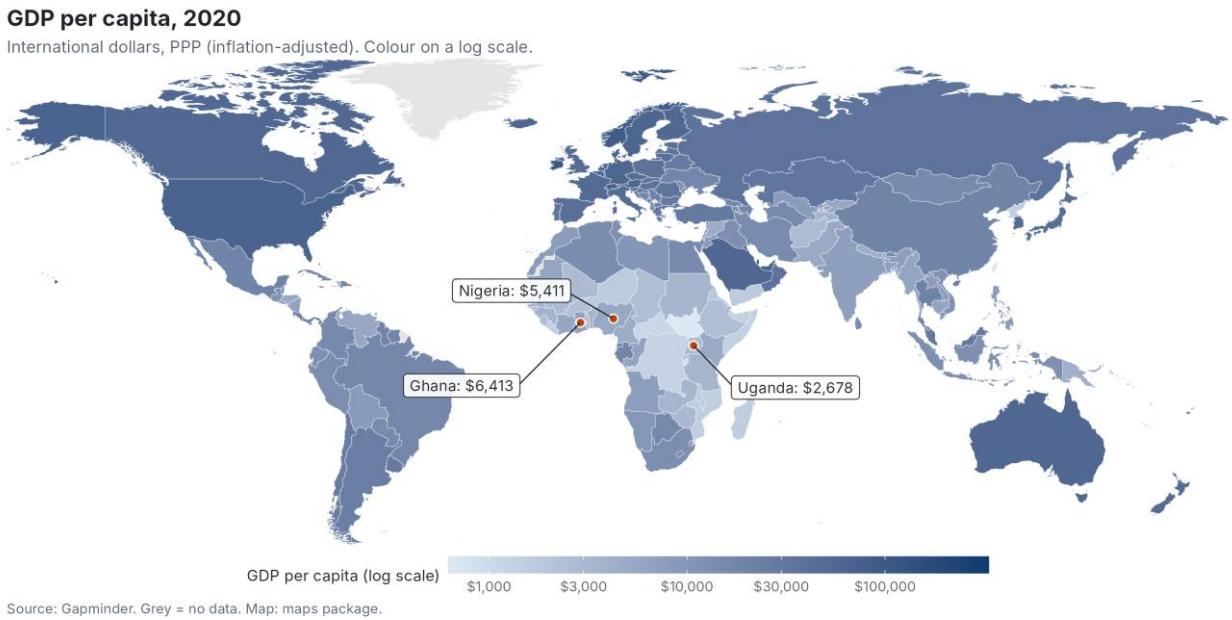


Figure 1. GDP per capita by country, 2020 (log colour scale). Labels show actual values for the focus countries.

1.3. 3. Life expectancy, 2020

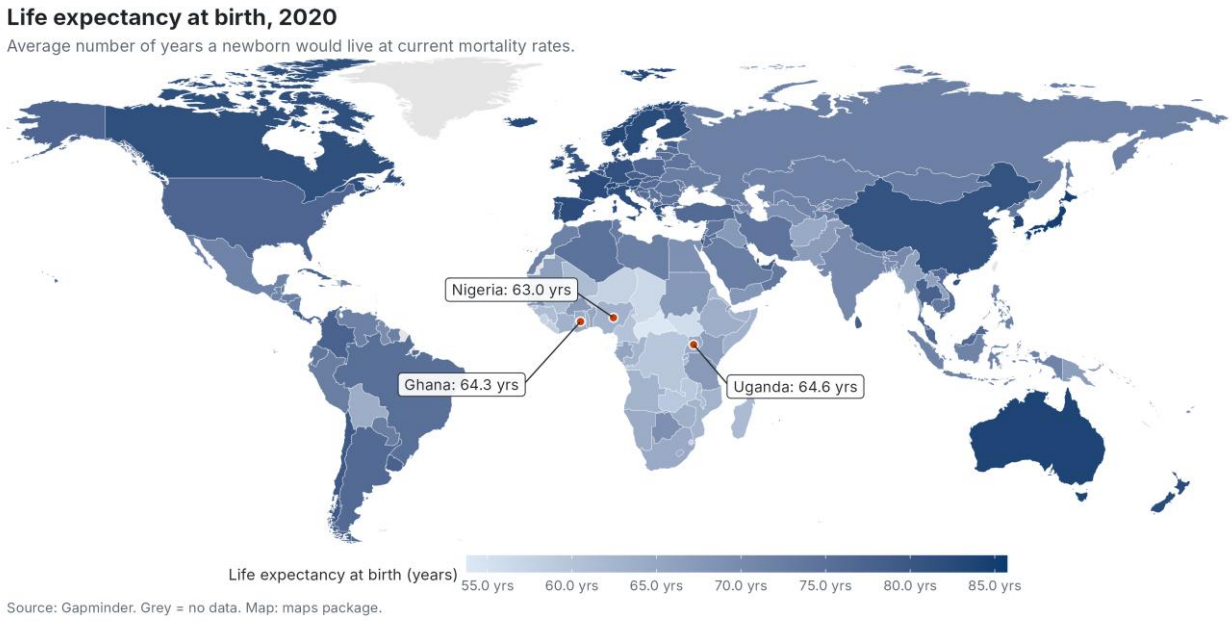


Figure 2. Life expectancy at birth by country, 2020. Labels show actual values for the focus countries.

1.4. 4. Relationship between the two indicators

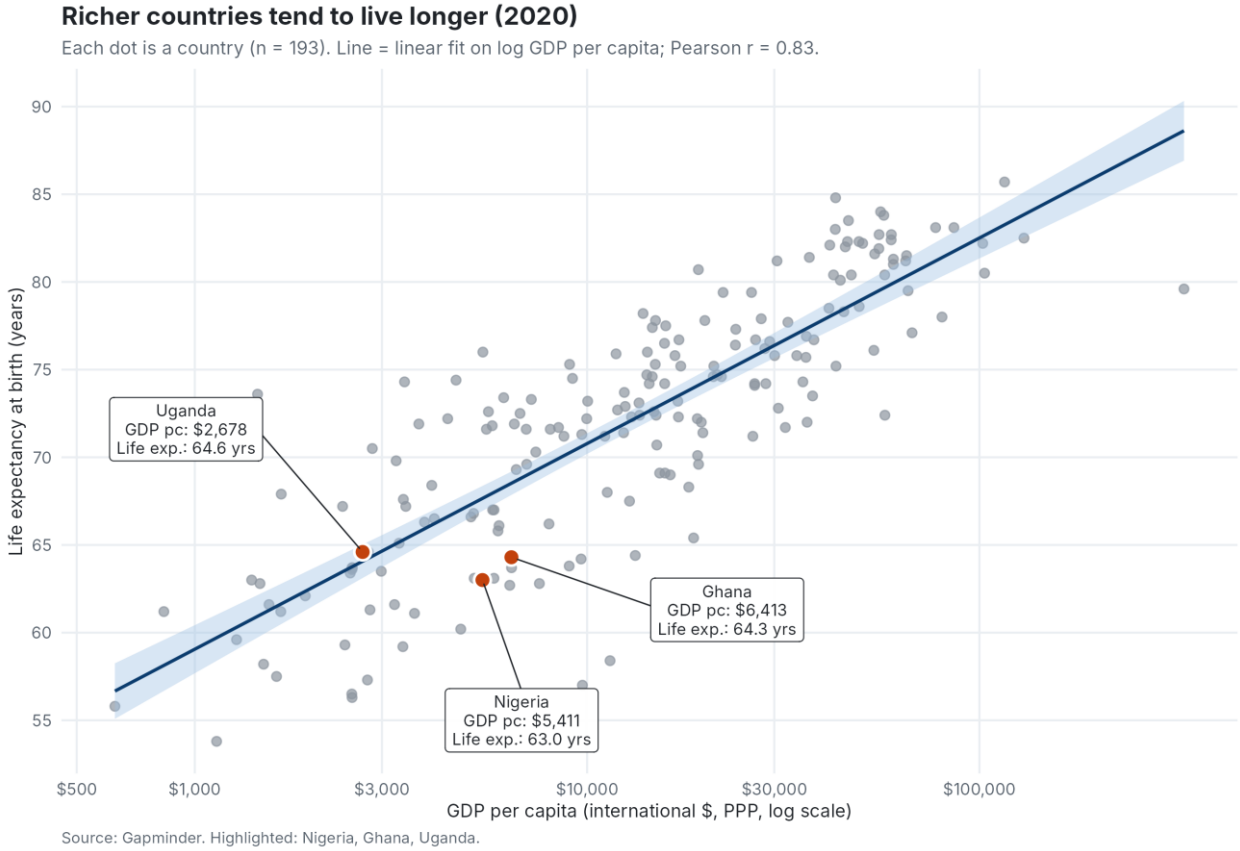


Figure 3. Life expectancy against GDP per capita (log scale), 2020, with a linear fit and 95% confidence band.

1.5. 5. Focus countries

Country	GDP per capita (int. $, PPP)	Life expectancy (years)	Predicted by trend (years)	Above/below trend (years)
Nigeria	$5,411	63.0	67.7	-4.7
Ghana	$6,413	64.3	68.5	-4.2
Uganda	$2,678	64.6	64.1	+0.5
World median (all countries)	**$14,397**	**72.6**		

Nigeria: GDP per capita of $5,411 (rank 147 of 193) and life expectancy of 63.0 yrs (rank 168 of 193). The fitted line predicts 67.7 yrs for a country at this income level, so Nigeria is 4.7 years below the trend.
Ghana: GDP per capita of $6,413 (rank 135 of 193) and life expectancy of 64.3 yrs (rank 157 of 193). The fitted line predicts 68.5 yrs for a country at this income level, so Ghana is 4.2 years below the trend.
Uganda: GDP per capita of $2,678 (rank 172 of 193) and life expectancy of 64.6 yrs (rank 153 of 193). The fitted line predicts 64.1 yrs for a country at this income level, so Uganda is 0.5 years above the trend.

1.6. 6. Notes
GDP per capita is in inflation-adjusted international dollars at purchasing power parity, as supplied by Gapminder; it is plotted on a log scale because the relationship with life expectancy is roughly linear in log income. Recent Gapminder values can include estimates or projections. 2020 life expectancy may also reflect early COVID-19 effects. Correlation does not establish causation: income, health spending, education and disease burden are interlinked.

2. Reflection of the analysis

2.1 My personal reflection of the analysis.

From the data set, life expectancy tends to increase with increasing GDP per capita. Also, the map shows that Africa has the lowest GDP per capita and the lowest life expectancy, which is concerning.

2.2 My Reflection about using AI for this analysis

The analysis was completed remarkably quickly, much faster than I could have done it myself. I had specifically instructed the AI to conduct the analysis using R, because I have done some analysis in R many years ago, and it was interesting to see how efficiently it could carry out the task.

Personally, I would be more comfortable allowing AI to conduct an entire analysis that I already know how to perform manually. This is because I would have a better understanding of the analytical process, be able to assess the accuracy of the outputs, identify and correct errors, and make adjustments where necessary. For me, AI should make tasks easier and more efficient, not replace human expertise and judgement.

Following this AI course, I intend to strengthen my data analysis skills further, particularly in R, by practicing how to perform this analysis (the map development) and other analytical tasks independently. My goal is to understand the underlying methods and logic well enough to confidently interpret and validate the results produced by AI.

Once I have developed that level of competence, I will be more comfortable delegating entire analytical tasks to AI. I believe this will allow me to combine my own technical knowledge and contextual understanding with AI's speed and efficiency. More importantly, I will be better positioned to give AI precise instructions, provide the necessary context, and guide it through more nuanced and complex analyses, while maintaining ownership of the process and confidence in the final outputs.

Ultimately, I see AI as a powerful tool that can enhance my capabilities as a researcher. The more I understand the work myself, the better I will be able to use AI effectively, critically evaluate its outputs, and harness its potential to improve the quality and efficiency of my research.